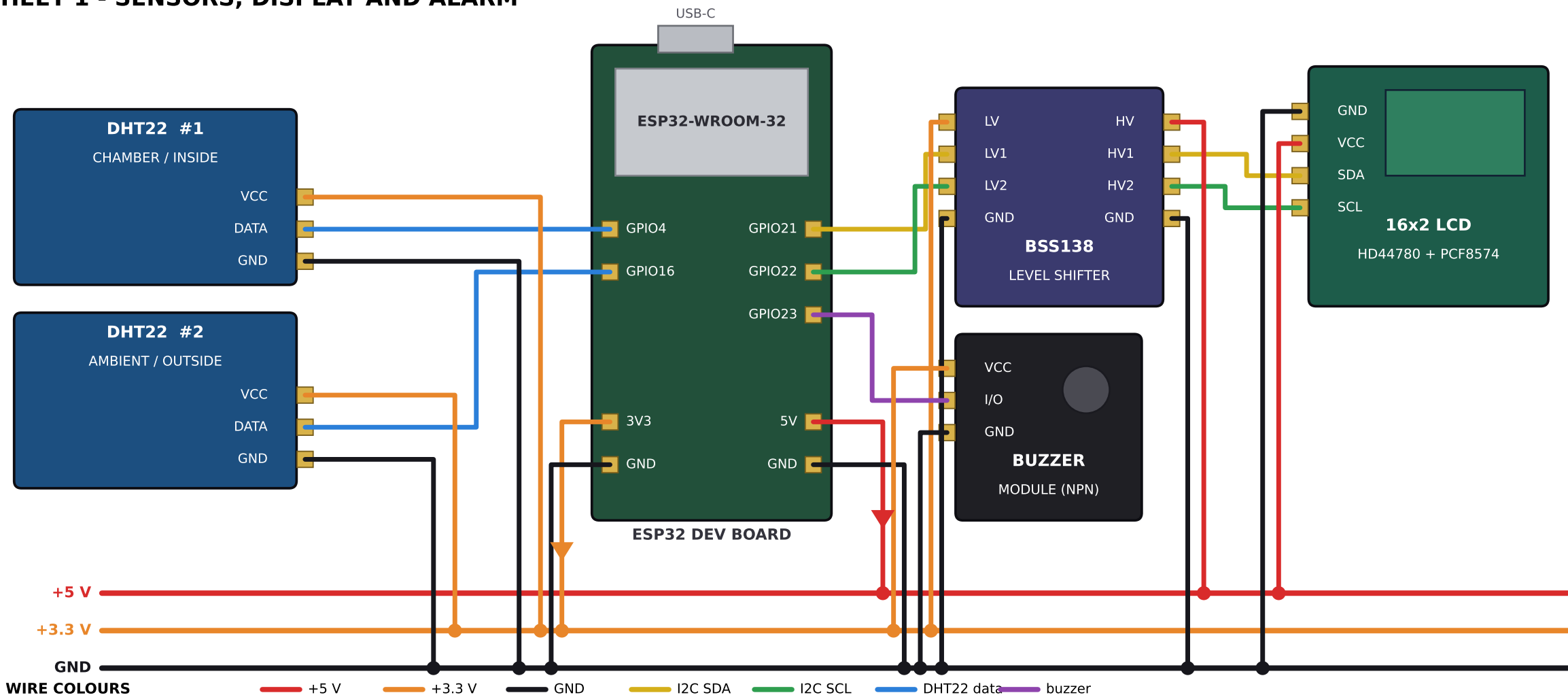


SHEET 1 - SENSORS, DISPLAY AND ALARM



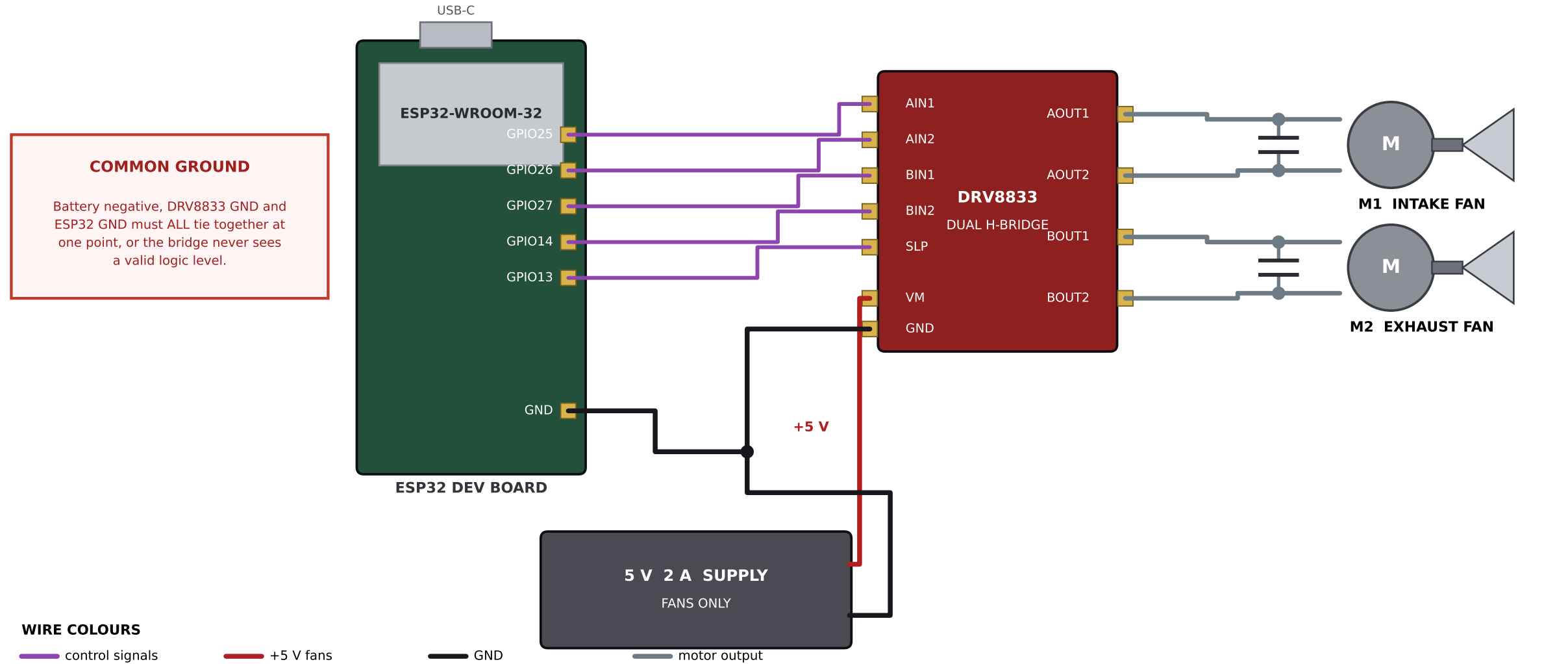
Wires that cross are NOT connected unless a solid dot is shown.
Pin headers are grouped for clarity - match pins by silkscreen LABEL, not position.
The LCD is the only 5 V part: its SDA/SCL MUST pass through the BSS138.

3V3 and 5V are OUTPUTS of the dev board

USB feeds the board and its regulator makes 3.3 V - both pins SOURCE the rails. NEVER feed an external supply into 3V3: it bypasses the regulator and destroys the ESP32.

ESDL PROJECT 1 - WIRING DIAGRAM		
Sensors, Display and Alarm		
TJ / Tabitha	2026-09-16	REV A SH 1/3

SHEET 2 - MOTOR DRIVE AND POWER



ESDL PROJECT 1 - WIRING DIAGRAM		
Motor Drive and Power		
TJ / Tabitha	2026-09-16	REV A SH 2/3

Fan supply NEVER comes from the ESP32 or USB. Brushless fans need no suppression caps - that risk is designed out.

SHEET 3 - CONNECTION TABLE

Tick each row as you wire it. Build one component at a time and verify with the serial console before the next.

	COMPONENT	PIN	CONNECTS TO	WIRE	NOTE
☐	DHT22 #1 (chamber)	VCC	+3.3 V rail	orange	module has the pull-up
☐		DATA	ESP32 GPIO 4	blue	3.3 V logic, direct
☐		GND	GND rail	black	
☐	DHT22 #2 (ambient)	VCC	+3.3 V rail	orange	keep clear of exhaust plume
☐		DATA	ESP32 GPIO 16	blue	3.3 V logic, direct
☐		GND	GND rail	black	
☐	BSS138 shifter	LV	+3.3 V rail	orange	low side reference
☐		LV1	ESP32 GPIO 21	yellow	SDA, 3.3 V side
☐		LV2	ESP32 GPIO 22	green	SCL, 3.3 V side
☐		HV	+5 V rail	red	high side reference
☐	16x2 LCD (PCF8574)	HV1	LCD SDA	yellow	5 V side
☐		HV2	LCD SCL	green	5 V side
☐		GND x2	GND rail	black	both GND pins
☐		VCC	+5 V rail	red	5 V for contrast
☐		GND	GND rail	black	
☐		SDA / SCL	BSS138 HV1 / HV2	yellow / green	NEVER direct to ESP32
☐					
☐	Buzzer module	VCC	+3.3 V rail	orange	module with onboard NPN
☐		I/O	ESP32 GPIO 23	purple	
☐		GND	GND rail	black	
☐	DRV8833 driver	AIN1 / AIN2	ESP32 GPIO 25 / 26	purple	intake fan
☐		BIN1 / BIN2	ESP32 GPIO 27 / 14	purple	exhaust fan
☐		SLP (nSLEEP)	ESP32 GPIO 13	purple	HIGH enables the bridge
☐		VM	5 V fan supply +	red	never from the ESP32
☐	Power	GND	Battery - AND ESP32 GND	black	COMMON GROUND - mandatory
☐		AOUT1 / AOUT2	M1 intake fan	grey	brushless, one direction
☐		BOUT1 / BOUT2	M2 exhaust fan	grey	brushless, one direction
☐		ESP32 3V3	+3.3 V rail	orange	logic supply
☐		ESP32 5V	+5 V rail	red	USB-derived, LCD only
☐		ESP32 GND	GND rail	black	

ESDL PROJECT 1 - WIRING DIAGRAM		
Connection Table		
TJ / Tabitha	2026-09-16	REV A SH 3/3